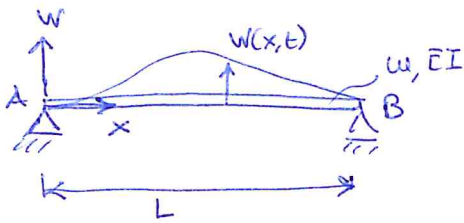
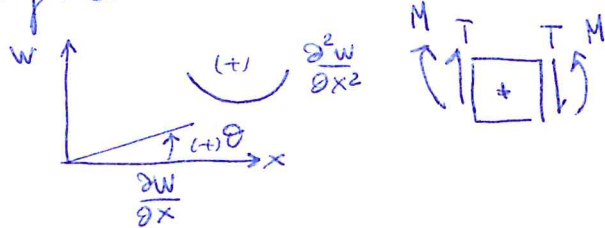


BENDING VIBRATIONS OF BEAMS

A) SIMPLY SUPPORTED BEAM



sign convention:



$$\text{hinge in A} \rightarrow \begin{cases} w(0,t) = 0 \\ M(0,t) = 0 \end{cases}$$

$$\text{hinge in B} \rightarrow \begin{cases} w(L,t) = 0 \\ M(L,t) = 0 \end{cases}$$

$$\begin{aligned} w(x,t) &= [A \cos \gamma x + B \sin \gamma x + C \cosh \gamma x + D \sinh \gamma x] \cos(\omega t + \varphi) \\ w'(x,t) &= \gamma [-A \sin \gamma x + B \cos \gamma x + C \sinh \gamma x + D \cosh \gamma x] \cos(\omega t + \varphi) \\ w''(x,t) &= \gamma^2 [-A \cos \gamma x - B \sin \gamma x + C \cosh \gamma x + D \sinh \gamma x] \cos(\omega t + \varphi) \\ w'''(x,t) &= \gamma^3 [-A \sin \gamma x + B \cos \gamma x + C \sinh \gamma x + D \cosh \gamma x] \cos(\omega t + \varphi) \end{aligned}$$

$$\text{beef } w'(x,t) = \frac{\partial w}{\partial x}; \quad w'' = \frac{\partial^2 w}{\partial x^2}; \quad w''' = \frac{\partial^3 w}{\partial x^3}$$

$$M(x,t) = EI \frac{\partial^2 w}{\partial x^2}; \quad T = EI \frac{\partial^3 w}{\partial x^3}; \quad \vartheta = \frac{\partial w}{\partial x}$$

$$\gamma^4 = \omega^2 \frac{m}{EI} \quad \begin{matrix} m \text{ [kg/m]} \\ I \text{ [m}^4\text{]} \end{matrix}$$

$$\begin{cases} [A + C] \cos(\omega t + \varphi) = 0 \\ EI \gamma^2 [-A + C] \cos(\omega t + \varphi) = 0 \\ [A \cos \gamma L + B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0 \\ EI \gamma^2 [-A \cos \gamma L - B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0 \end{cases}$$

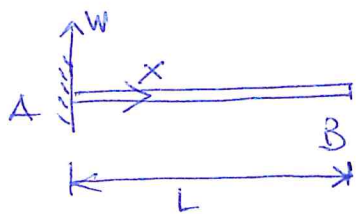
$$\underline{x} = \{A \ B \ C \ D\}^T$$

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ \cos \gamma L & \sin \gamma L & \cosh \gamma L & \sinh \gamma L \\ -\cos \gamma L & -\sin \gamma L & \cosh \gamma L & \sinh \gamma L \end{bmatrix}$$

$$\rightarrow \det(H) = 0 \rightarrow \gamma_i = i = 1 \dots N$$

$$\rightarrow \omega_i = \gamma^2 \frac{EI}{m} \rightarrow f_i = \frac{\omega_i}{2\pi}$$

B) CANTILEVER

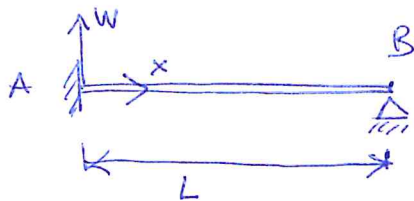


$$\begin{aligned} \text{clump in A} &\rightarrow \begin{cases} w(0, t) = 0 \\ \theta(0, t) = 0 \end{cases} \\ \text{free end in B} &\rightarrow \begin{cases} M(L, t) = 0 \\ T(L, t) = 0 \end{cases} \end{aligned}$$

$$\begin{cases} [A + C] \cos(\omega t + \varphi) = 0 \\ \cancel{[B + D] \cos(\omega t + \varphi) = 0} \\ \cancel{[-A \cos \gamma L - B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0} \\ \cancel{[A \sin \gamma L - B \cos \gamma L + C \sinh \gamma L + D \cosh \gamma L] \cos(\omega t + \varphi) = 0} \end{cases}$$

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos \gamma L & -\sin \gamma L & \cosh \gamma L & \sinh \gamma L \\ \sin \gamma L & -\cos \gamma L & \sinh \gamma L & \cosh \gamma L \end{bmatrix} = 0 \rightarrow \det(H) = 0 \rightarrow \gamma_i, i=1 \dots N$$

C) CLAMP + HINGE

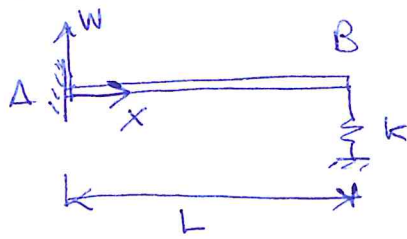


$$\begin{aligned} \text{clump in A} &\rightarrow \begin{cases} w(0, t) = 0 \\ \theta(0, t) = 0 \end{cases} \\ \text{hinge in B} &\rightarrow \begin{cases} w(L, t) = 0 \\ M(L, t) = 0 \end{cases} \end{aligned}$$

$$\begin{cases} [A + C] \cos(\omega t + \varphi) = 0 \\ \cancel{[B + D] \cos(\omega t + \varphi) = 0} \\ [A \cos \gamma L + B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0 \\ \cancel{[-A \cos \gamma L - B \sin \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0} \end{cases}$$

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \cos \gamma L & \sin \gamma L & \cosh \gamma L & \sinh \gamma L \\ -\cos \gamma L & -\sin \gamma L & \cosh \gamma L & \sinh \gamma L \end{bmatrix} = 0 \rightarrow \det(H) = 0 \rightarrow \gamma_i, i=1, N$$

1) CLAMP + SPRING



$$\text{clump in A} \rightarrow \begin{cases} w(0,t) = 0 \\ \theta(0,t) = 0 \end{cases}$$

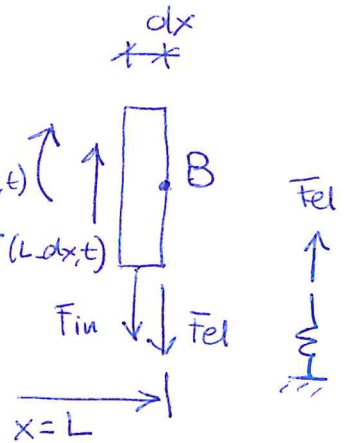
free end with spring in B:

$$\sum M_B = 0 : M(L-dx, t) + T(L-dx, t) dx - F_{in} dx/2 = 0$$

negligible because higher order infinitesimal with respect to M

$$M(L-dx, t) = M(L, t) - \frac{\partial M}{\partial x} \bigg|_{x=L} dx = 0$$

higher order infinitesimal



$$\sum \bar{F} = 0 : T(L-dx, t) - F_{el} - m \frac{\partial^2 w}{\partial t^2} \bigg|_{x=L-dx/2} dx = 0$$

$$T(L, t) - \frac{\partial T}{\partial x} \bigg|_{x=L} dx - kW(L, t) = 0$$

higher order infinitesimal

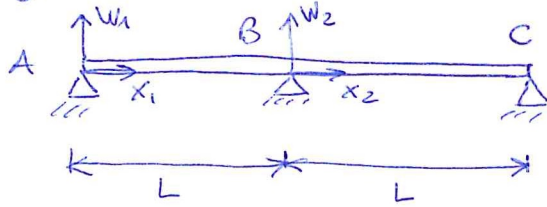
$$\boxed{T(L, t) - kW(L, t) = 0}$$

$$\begin{cases} [A+C] \cos(\omega t + \varphi) = 0 \\ [B+D] \cos(\omega t + \varphi) = 0 \\ EI \delta^2 [-A \cos \gamma L - B \sinh \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0 \\ EI \delta^3 [A \sinh \gamma L - B \cos \gamma L + C \sinh \gamma L + D \cosh \gamma L] \cos(\omega t + \varphi) - [A \cos \gamma L + B \sinh \gamma L + C \cosh \gamma L + D \sinh \gamma L] \cos(\omega t + \varphi) = 0 \end{cases}$$

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos \gamma L & -\sinh \gamma L & \cosh \gamma L & \sinh \gamma L \\ \left(\sinh \gamma L - \frac{k}{EI \gamma^3} \cos \gamma L \right) & \left(-\cos \gamma L - \frac{k}{EI \gamma^3} \sinh \gamma L \right) & \left(\sinh \gamma L - \frac{k}{EI \gamma^3} \cosh \gamma L \right) & \left(\cosh \gamma L - \frac{k}{EI \gamma^3} \sinh \gamma L \right) \end{bmatrix}$$

$$\det(H) = 0 \rightarrow \gamma_i ; i = 1 \dots N$$

E) CONTINUOUS BEAM

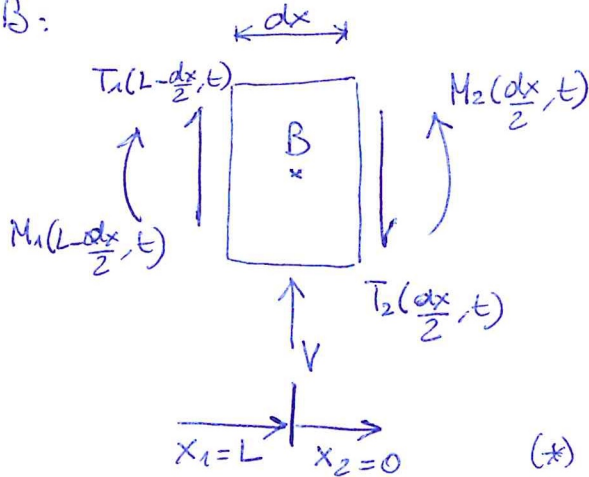


• discontinuity in B → end of the w_1 function and beginning of a new w_2 function

hinge in A →
$$\begin{cases} w_1(0,t) = 0 \\ M_1(0,t) = 0 \end{cases}$$

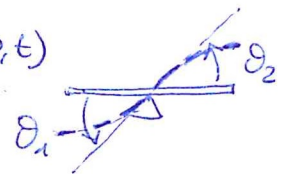
hinge in C →
$$\begin{cases} w_2(L,t) = 0 \\ M_2(L,t) = 0 \end{cases}$$

pin in B:



$$\begin{cases} w_1(L,t) = 0 \\ w_2(0,t) = 0 \end{cases} (*)$$

$$\begin{cases} \theta_1(L,t) = \theta_2(0,t) \\ \sum M_B = 0 \end{cases}$$



(*) one of the two can be substituted with $w_1(L,t) = w_2(0,t)$

$\sum M_B = 0:$
$$M_1(L - \frac{dx}{2}, t) + T_1(L - \frac{dx}{2}, t) \frac{dx}{2} + T_2(\frac{dx}{2}, t) \frac{dx}{2} - M_2(\frac{dx}{2}, t) = 0$$

higher order infinitesimal

$$M_1(L,t) - \frac{\partial M_1}{\partial x} \Big|_{L} \frac{dx}{2} - M_2(0,t) + \frac{\partial M_2}{\partial x} \Big|_0 \frac{dx}{2} = 0 \rightarrow \boxed{M_1(L,t) - M_2(0,t) = 0}$$

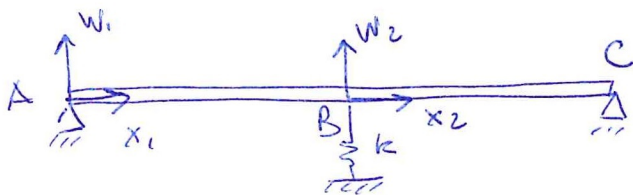
$$\begin{cases} w_1(x_1,t) = [A_1 \cosh \gamma x_1 + B_1 \sinh \gamma x_1 + C_1 \cosh \gamma x_1 + D_1 \sinh \gamma x_1] \cos(\omega t + \varphi) \\ w_2(x_2,t) = [A_2 \cosh \gamma x_2 + B_2 \sinh \gamma x_2 + C_2 \cosh \gamma x_2 + D_2 \sinh \gamma x_2] \cos(\omega t + \varphi) \end{cases}$$

$$\begin{cases} [A_1 + C_1] \cos(\omega t + \varphi) = 0 \\ EI \gamma^2 [-A_1 + C_1] \cos(\omega t + \varphi) = 0 \\ [A_1 \cosh \gamma L + B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sinh \gamma L] \cos(\omega t + \varphi) = 0 \\ [A_2 + C_2] \cos(\omega t + \varphi) = 0 \\ \cancel{\gamma} [-A_1 \sinh \gamma L + B_1 \cosh \gamma L + C_1 \sinh \gamma L + D_1 \cosh \gamma L] \cos(\omega t + \varphi) - \cancel{\gamma} [B_2 + D_2] \cos(\omega t + \varphi) = 0 \\ EI \gamma^4 [-A_1 \cosh \gamma L - B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sinh \gamma L] \cos(\omega t + \varphi) - \cancel{\gamma^2} EI [-A_2 + C_2] \cos(\omega t + \varphi) = 0 \\ [A_2 \cosh \gamma L + B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sinh \gamma L] \cos(\omega t + \varphi) = 0 \\ EI \gamma^2 [-A_2 \cosh \gamma L - B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sinh \gamma L] \cos(\omega t + \varphi) = 0 \end{cases}$$

$$[H(\omega)] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \cos \omega L & \sin \omega L & \cosh \omega L & \sinh \omega L & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ -\sin \omega L & \cos \omega L & \sinh \omega L & \cosh \omega L & 0 & -1 & 0 & -1 \\ -\cos \omega L & -\sin \omega L & \cosh \omega L & \sinh \omega L & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & \cos \omega L & \sin \omega L & \cosh \omega L & \sinh \omega L \\ 0 & 0 & 0 & 0 & -\cos \omega L & -\sin \omega L & \cosh \omega L & \sinh \omega L \end{bmatrix} \rightarrow \det(H) = 0$$

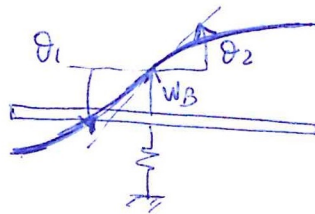
$$\underline{x} = \{A_1 \ B_1 \ C_1 \ D_1 \ A_2 \ B_2 \ C_2 \ D_2\}^T$$

F) CONTINUOUS BEAM SUPPORTED BY A SPRING

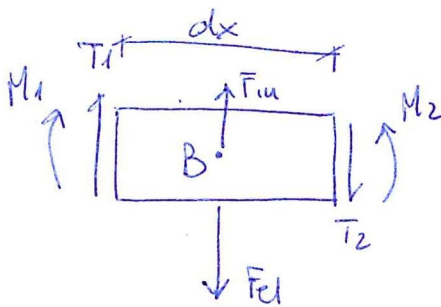


$$\begin{aligned} \text{hinge in A: } & \begin{cases} w_1(0, t) = 0 \\ M_1(0, t) = 0 \end{cases} \\ \text{hinge in C: } & \begin{cases} w_2(L, t) = 0 \\ M_2(L, t) = 0 \end{cases} \end{aligned}$$

SPRING in B:



$$\begin{aligned} w_B &= w_1(L, t) = w_2(0, t) \\ \theta_1(L, t) &= \theta_2(0, t) \end{aligned} \quad \text{eq. 3 and 4}$$



$$\bar{F}_m = m dx \frac{\partial^2 w_2}{\partial x_2^2} \bigg|_0 : \text{infinitesimal Term}$$

$$\sum M_B = 0 : M_1 + \cancel{T_1 \frac{dx}{2}} + \cancel{T_2 \frac{dx}{2}} - M_2 = 0$$

infinitesimal

$$x_1 = L \quad x_2 = 0$$

$$M_1(L, t) - \cancel{\frac{\partial M_1}{\partial x_1} \bigg|_L \frac{dx}{2}} - M_2(0, t) - \cancel{\frac{\partial M_2}{\partial x_2} \bigg|_0 \frac{dx}{2}} = 0$$

$$\rightarrow \boxed{M_1(L, t) - M_2(0, t) = 0} \quad \text{eq. 5}$$

$$\sum F = 0 :$$

$$T_1 - F_{el} - T_2 = 0$$

$$\boxed{T_1(L, t) - \cancel{\frac{\partial T_1}{\partial x_1} \bigg|_L \frac{dx}{2}} - T_2(0, t) - \cancel{\frac{\partial T_2}{\partial x_2} \bigg|_0 \frac{dx}{2}} - k w_2(0, t) = 0} \quad \text{eq. 6}$$

$$[A_1 + C_1] \cos(\omega t + \varphi) = 0$$

$$EI \gamma^2 [-A_1 + C_1] \cos(\omega t + \varphi) = 0$$

$$[A_1 \cos \gamma L + B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sin \gamma L] \cos(\omega t + \varphi) - [A_2 + C_2] \cos(\omega t + \varphi) = 0$$

$$[-A_1 \sinh \gamma L + B_1 \cos \gamma L + C_1 \sin \gamma L + D_1 \cosh \gamma L] \cos(\omega t + \varphi) - [B_1 + D_1] \cos(\omega t + \varphi) = 0$$

$$EI \gamma^2 [-A_1 \cos \gamma L - B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sin \gamma L] \cos(\omega t + \varphi) - EI \gamma^2 [-A_2 + C_2] \cos(\omega t + \varphi) = 0$$

$$EI \gamma^3 [A_1 \sinh \gamma L - B_1 \cos \gamma L + C_1 \sin \gamma L + D_1 \cosh \gamma L] \cos(\omega t + \varphi) - EI \gamma^3 [-B_2 + D_2] \cos(\omega t + \varphi) +$$

$$[A_2 \cos \gamma L + B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sin \gamma L] \cos(\omega t + \varphi) - k [A_2 + C_2] \cos(\omega t + \varphi) = 0$$

$$EI \gamma^2 [-A_2 \cos \gamma L - B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sin \gamma L] \cos(\omega t + \varphi) = 0$$

$$[H(\gamma)] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \cos \gamma L & \sinh \gamma L & \cosh \gamma L & \sin \gamma L & -1 & 0 & -1 & 0 \\ -\sinh \gamma L & \cos \gamma L & \sin \gamma L & \cosh \gamma L & 0 & -1 & 0 & -1 \\ -\cos \gamma L & -\sinh \gamma L & \cosh \gamma L & \sin \gamma L & 1 & 0 & -1 & 0 \\ \sinh \gamma L & -\cos \gamma L & \sin \gamma L & \cosh \gamma L & -k/EI \gamma^3 & 1 & -k/EI \gamma^3 & -1 \\ 0 & 0 & 0 & 0 & \cos \gamma L & \sinh \gamma L & \cosh \gamma L & \sin \gamma L \\ 0 & 0 & 0 & 0 & -\cos \gamma L & -\sinh \gamma L & \cosh \gamma L & \sin \gamma L \end{bmatrix} \rightarrow \det(H) = 0$$